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F															
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM DRC BUS PT H05															
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV	
				Prepared		YR		MV Switchgear for MNA Serang Plant							
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F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet		Scale		= H05	F
					Prepared	YR	MV Switchgear for MNA Serang Plant				MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Customer	PT Multimas Nabati Asahan	& 308	
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A	
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
B	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						B	
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
C	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						C	
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
D	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK						D	
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
E	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)						E	
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	REFERENCE DESIGNATIONS MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Scale			= H05	F
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

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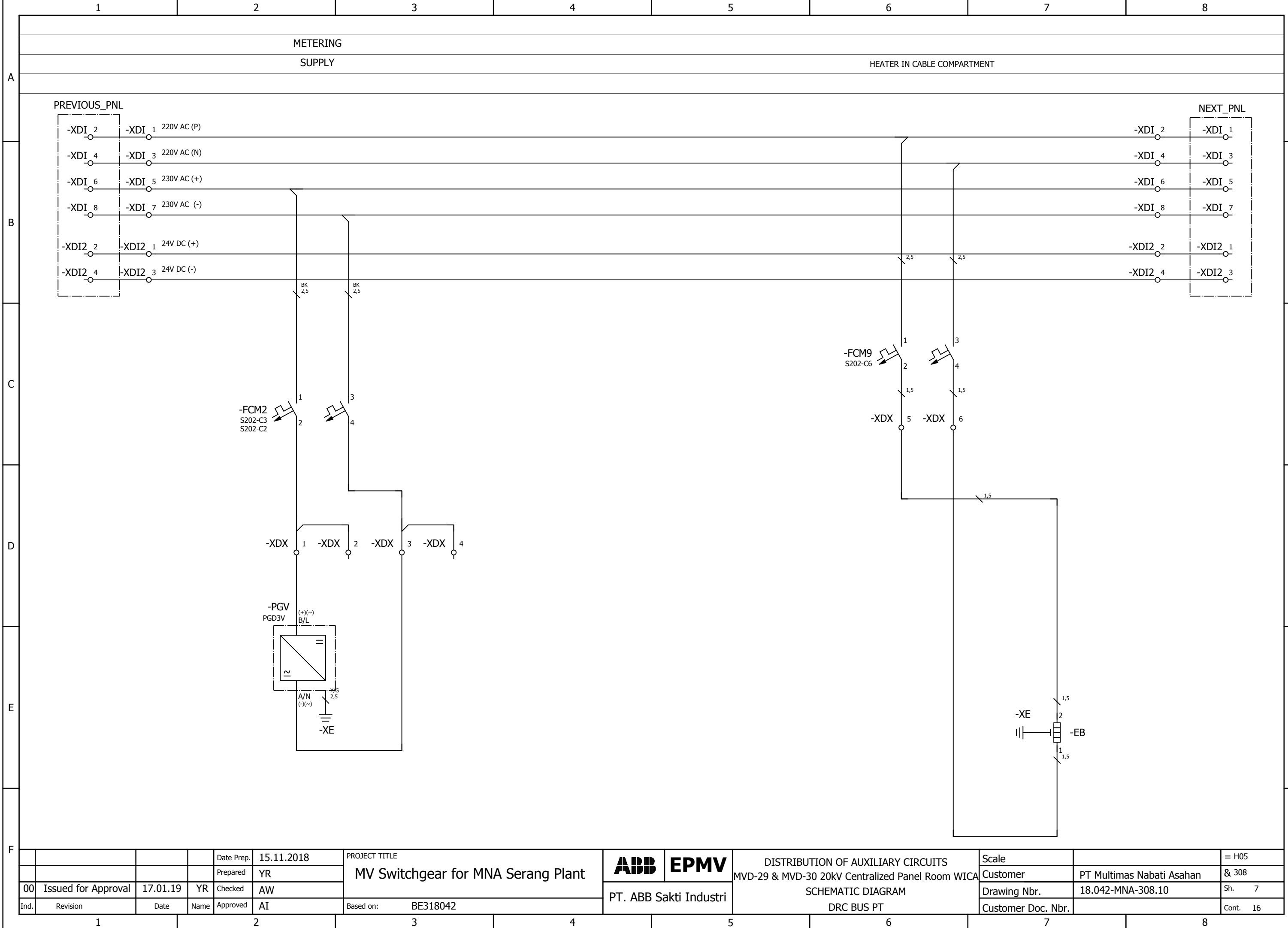
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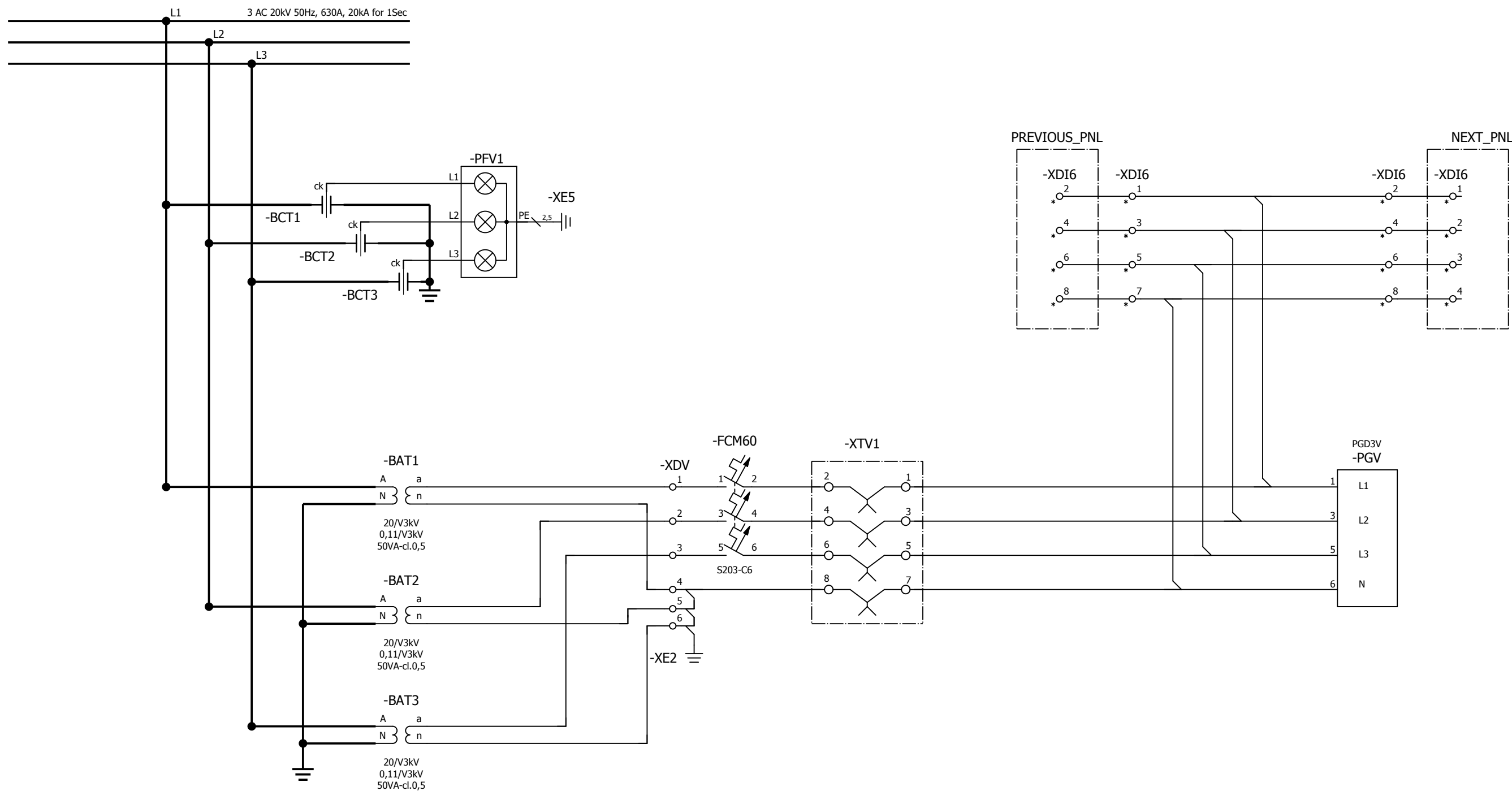
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A	-TFA	ACTIVE POWER TRANSDUCERS	-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	A
	-TFC	CURRENT TRANSDUCERS		-XD12	
B	-TFF	FREQUENCY TRANSDUCERS	-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION	B
	-TFJ	ACTIVE ENERGY TRANSDUCERS		-XD14	
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS	
	-TFM	MULTIFUNCTION TRANSDUCERS	-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS	
	-TFP	POWER-FACTOR TRANSDUCERS	-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS	
	-TFQ	REACTIVE POWER TRANSDUCERS	-XD19	TERMINAL BLOCK FOR SPECIAL USE	
	-TFS	SIGNAL CONVERTERS	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD	
	-TFV	VOLTAGE TRANSDUCERS	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS	
	-WA	BUSBARS	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT	
	C	-WBC	POWER CABLES	-XDS	
-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB	
-WGC		CONTROL CABLES	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
-WH		OPTICAL FIBRES	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS	
-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1	
-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2	
-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3	
-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR	
-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDX	SUPPORT TERMINAL BLOCK	
D	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE	D
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE	
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE	
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE	
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE	
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE	
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT	
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR			
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			
	-XDC	CUSTOMER TERMINAL BLOCK			
E	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			E
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR			
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS			
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE			
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS			
F	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR			F
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS			
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS			
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)			
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)			

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00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-308.10	Sh. 6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				DRC BUS PT	Customer Doc. Nbr.		Cont. 16

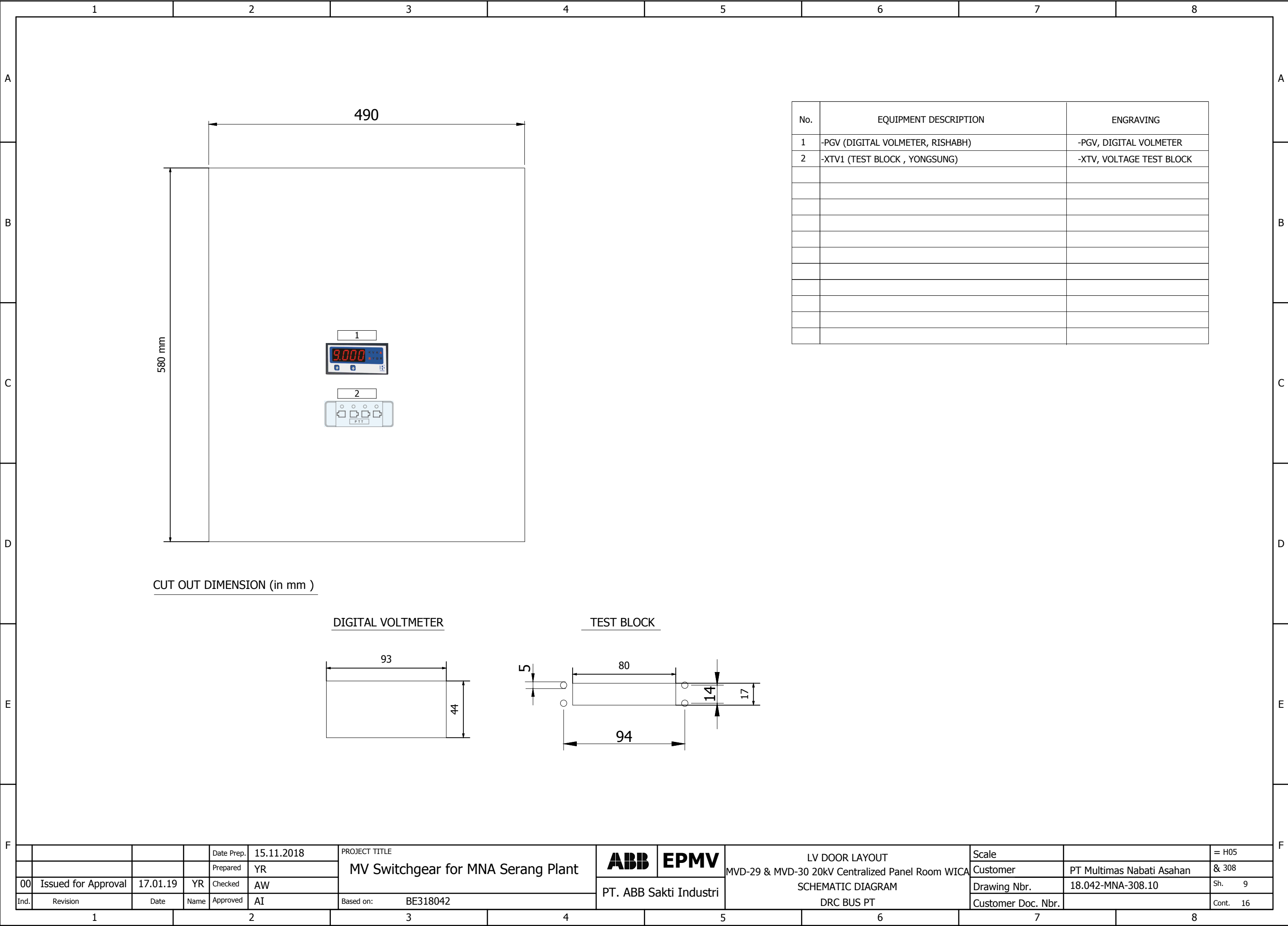
1	2	3	4	5	6	7	8
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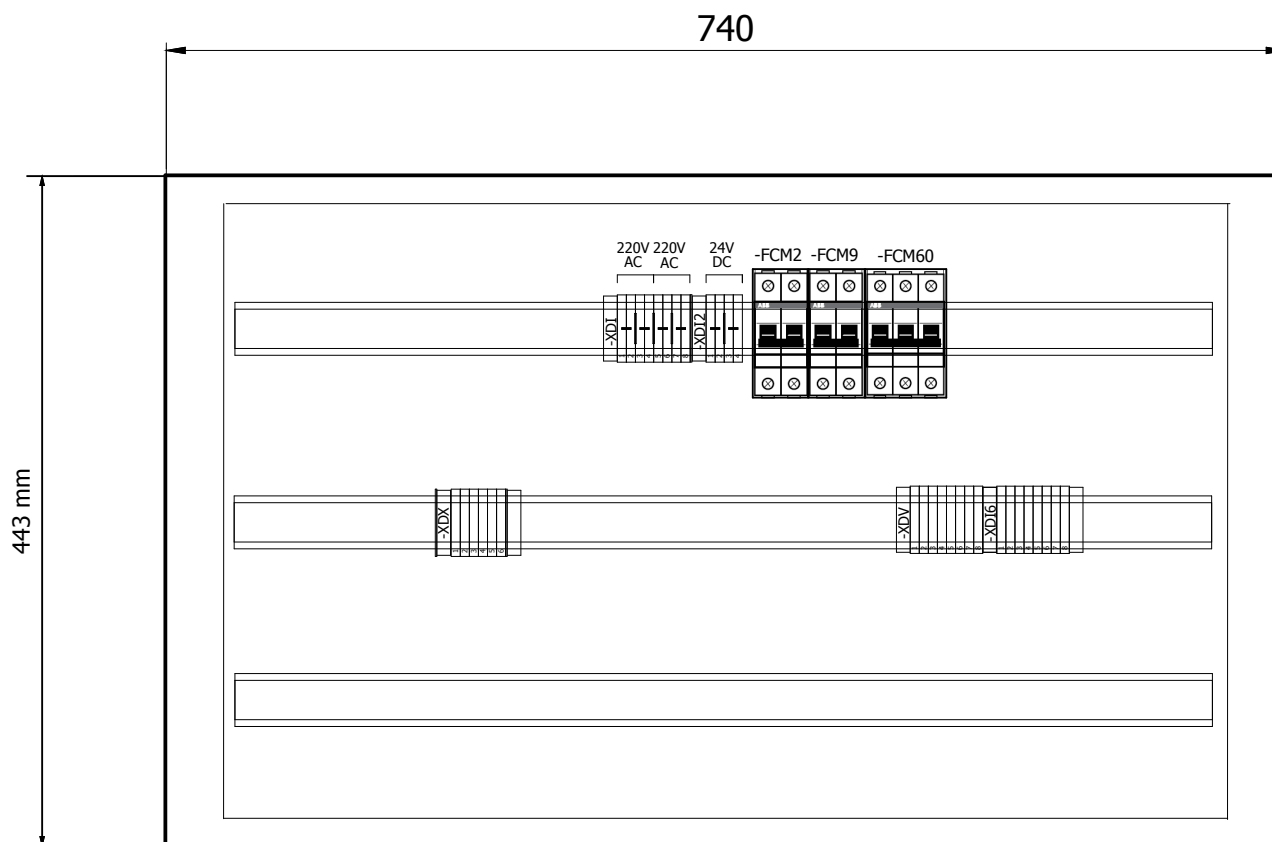
[illegible]



			Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			MAIN CIRCUITS MVD-29 & MVD-30 20kV Centralized Panel Room WICAP SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05
			Prepared	YR					Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	17.01.19	YR	Checked		AW	Drawing Nbr.		18.042-MNA-308.10	Sh.	8
Ind.	Revision	Date	Name	Approved	Aİ	Based on:	BE318042		Customer Doc. Nbr.		Cont. 16
						PT. ABB Sakti Industri					

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[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05
				Prepared	YR							Customer
00	Issued for Approval	17.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-308.10	Sh. 10
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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Summarized parts list																IDABB_F02_005-IDABB			
																Carry over: 0.00			
TAG ID				DESCRIPTION						MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-EB				HEATER 230VAC, 50W						SANYANG		CDXN010033P0009		1		/7.7:E			
-FCM2				MCB S202-C2 MCB. 2P, 2A						ABB		2CDS252001R0024		1		/7.2:C			
-FCM9				MCB S202-C6 MCB. 2P, 6A						ABB		2CDS252001R0064		1		/7.6:C			
-FCM60				MCB S203-C6 MCB. 3P, 6A						ABB		2CDS253001R0064		1		/8.4:C			
-PGV				Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM						RISHABH		DPM PGD3V		1		/7.2:D			
-XTV1				Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting						YONGSUNG		YSPTT-04C		1		/8.5:C			

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]